Lab 3 – Architecture Justification

Problem Statement #02: Automated Rubric Assignment Evaluator (ARAE) | PES1UG24CS189

Architecture Selection

We chose **Microservices Architecture** for the Automated Rubric Assignment Evaluator System.

The system is split into independently deployable services: Submission, Configuration, Peer Review, Results, Evaluation and an isolated Execution Sandbox. They are fronted by an API Gateway and connected through a Message Queue. This mirrors the five Jira epics from Lab 2.

Style	Assessment for ARAE	Verdict
Layered	Simple, but the evaluation engine cannot be scaled apart from the rest, and a crashing student program sits in the same deployable as results and configuration.	Not chosen
Client-Server	One central server is a bottleneck and single point of failure under 100 concurrent submissions.	Not chosen
Microservices	Independent scaling, fault isolation and a separate sandbox service match the workload and risk profile.	**Chosen**

Two Reasons for the Choice

- **Independent scaling for peak load (NFR-001, FR-001).** Almost all the heavy work is in evaluation, meaning unit tests run on student code. The Evaluation and Sandbox services can be scaled out on their own while lightweight services stay small. The queue buffers a burst of 100 concurrent uploads so no submission is dropped.
- **Fault isolation and independent change.** Student code can hang, crash or exhaust memory. In this style such a failure stays inside the sandbox service and cannot bring down Results, Peer Review or Configuration. Faculty-facing changes (rubric, weights, test suites, FR-004) can also be released without redeploying the evaluation engine.

Security Advantage

Untrusted code runs only in the Execution Sandbox Service, a separate container with no access to host files and no outbound network, which directly satisfies NFR-002. Since sandbox and data services are separate processes, a compromised submission cannot reach stored results or configuration. All external traffic passes through the API Gateway, which authenticates users and enforces role-based access, so a student can see only their own results (FR-005).

Performance Benefit

Evaluation is asynchronous. The Submission Service validates the ZIP (FR-002), publishes a job to the queue and returns immediately, and Evaluation workers pull jobs at their own pace. Adding workers raises throughput horizontally to keep results within the 10-second target (FR-001), and per-container memory limits keep the total footprint within the 1.5 GB budget under 100 concurrent submissions.

Trade-off Acknowledged

Microservices add operational complexity and network latency. For ARAE this is acceptable because the workload is naturally asynchronous (queue-based) and the isolation and scaling benefits outweigh the extra deployment effort.